Water Drop Method



Desktop app based on Tkinter to implement the Water Drop Method and determine the Sample Breakdown Energy of water-stable soil aggregates.

Distribution

The application is currently distributed as a compiled build for Windows 64-bit.

- GitHub Releases: https://github.com/alencina-faa/Water-drop-method/releases
- SourceForge: https://sourceforge.net/projects/water-drop-method/

Highlights

- GUI workflow with tabs for camera preview, threshold setup, measurement, drop energy, and video processing.
- Runtime state persisted per user on Windows.
- Package-safe import structure for src-layout projects.
- Basic automated test suite with structural and persistence checks.

Project Structure

- `src/Water_drop_method` : application package
- `tests` : automated tests
- `output` : local build artifacts (ignored by git)

Requirements

- Python 3.10+
- Dependencies:

- `pillow`
- `numpy`
- `matplotlib`
- `opencv-python`
- `nidaqmx` and/or `pyserial`

Install (Editable)

```
python -m venv .venv
.venv\Scripts\activate
pip install -e .
```

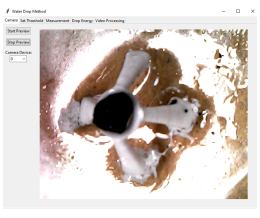
Run

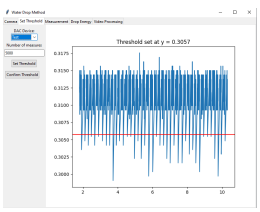
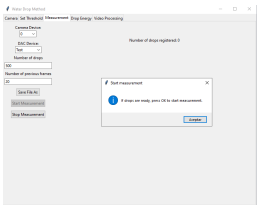
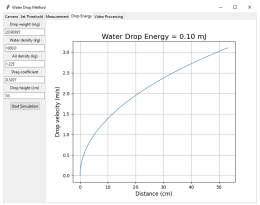
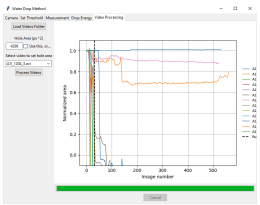
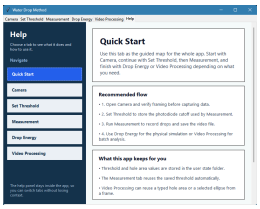
```
python -m Water_drop_method
```

Run Tests

```
pytest -q
```

Tabs Guide

Tab	What it is used for	Main actions	Screenshot
Camera	Preview the camera feed and validate framing before measurements.	Select camera device, Start Preview, Stop Preview.	

Tab	What it is used for	Main actions	Screenshot
Set Threshold	Compute and confirm the photodiode threshold.	Select DAC device, set number of measures, Set Threshold, Confirm Threshold.	
Measurement	Execute the drop measurement workflow.	Select camera and DAC, set drops and previous frames, Save File As, start/stop acquisition.	
Drop Energy	Estimate drop velocity and impact energy from physical parameters.	Configure physical values and run Start Simulation.	
Video Processing	Batch-process videos and analyze normalized area over frames.	Load videos folder, define hole area, Process Videos, inspect plots and outputs.	
Help	Browse a guided, clickable explanation for every tab inside the app.	Select a topic on the left, read the step-by-step guidance on the right.	

Runtime State Files

The app stores runtime state in a per-user writable folder:

- Windows: `%APPDATA%\\WaterDropMethod`
- Fallback: `<home>/WaterDropMethod`

Persisted files:

- `threshold.txt`
- `hole_area.txt`

Legacy Migration

If legacy `threshold.txt` or `hole_area.txt` files are found in the current working directory, the app performs a one-time best-effort copy to the per-user state folder when those values are requested.

These runtime files are intentionally excluded from version control.